Parth K Dubal

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Education

Purdue University **Aug 2024 – May 2028**

B.S. in Aeronautical and Astronautical Engineering, Minor in Business Economics *West Lafayette, IN*

- GPA: 3.81 / 4.00 | Dean's List (4x), Semester Honors (4x) | AAE Ambassador
- Relevant Coursework: Signals, Fluid Mechanics, Structures and Materials, Thermodynamics, Engineering Design, CAD Modeling in NX, Programming in C/Python/MATLAB

Experience & Technical Projects

Aerodynamics & Sizing Engineer – Flight Zero (Bio-Inspired UAV) **Aug 2026 – Present**

Purdue University *West Lafayette, IN*

- Developed conceptual sizing scripts in **MATLAB** to evaluate Maximum Takeoff Weight (MTOW) build-up and mission-segment energy balance for an autonomous, nature-inspired fixed-wing UAV.
- Conducted longitudinal trim drag analysis and stability estimations across operational regimes to optimize cruise efficiency and expand operability boundaries.
- Scripted **MATLAB** solvers for rudder/elevator sizing, aerodynamic loads (wing, fuselage, tail), and boom idealization.

Aerospace Engineering Intern – TarkX Defence Systems **May 2026 – Aug 2026**

*Project under the **Indian Navy*** *Bangalore, India*

- Developed a Python analytical thermal model for a diffuser nozzle, verifying transient results against **ANSYS Thermal** within a **1-second margin** to ensure thermal integrity.
- Down-selected optimal **Thrust Vector Control (TVC)** system; modeled mechanical assembly and conducted **ANSYS CFD simulations** to evaluate thrust vectoring and side forces.
- Compiled detailed **Bill of Materials (BOM)** by sizing and sourcing electro-mechanical actuators and flight-rated structural hardware.

Propulsion Engineer – Purdue Undergraduate Research Propulsion Lab **Aug 2025 – Aug 2026**

Purdue University *West Lafayette, IN*

- Engineered a Python **1-D transient thermal solver** (Bartz equation) coupled with **NASA CEA** to validate ∼15s burn limits and baseline 500 lbf thrust for a modular heatsink rocket engine.
- Characterized Secondary Injection Thrust Vector Control (SITVC) integration via Python scripts mapping shock boundary interactions, injection angles, and side force generation.

Apprentice Engineer – Materials and Manufacturing Research Laboratories **Dec 2025 – Present**

Purdue University *West Lafayette, IN*

- Authored SOP and Non-Conformance Report frameworks toward ISO 9001 Certification for Directed Energy Deposition (DED) and 5-axis CNC machining systems.
- Executed machine setup, calibration, and metal powder sieving/handling procedures for additive manufacturing workflows.

Career Lead – Aeronautical & Astronautical Engineering Student Council **Aug 2025 – Present**

AAESAC - Purdue University *West Lafayette, IN*

- Directing corporate outreach and operations for the Aerospace Career Exposition, hosting **10 major aerospace firms** (Lockheed Martin, General Atomics, United Airlines) for 1,000+ students.

Research

Undergraduate Researcher – PURT-UAVs: Biomedical Narcan Delivery **Aug 2024 – Sept 2025**

Purdue University *West Lafayette, IN*

- Designed an autonomous payload mechanism and modeled aerodynamic frameworks in **SolidWorks and Fusion 360**, reducing total airframe drag to increase top speed by **33% (45 → 60 mph)**.
- Awarded 1st Place Data Stewardship Award at the 2025 Spring Undergraduate Research Conference for flight telemetry collection.

Technical Skills

CAD & Modeling: SOLIDWORKS, Siemens NX, Creo Parametric, Fusion 360
Simulations & Analysis: ANSYS (Fluent, Mechanical, Thermal), XFLR5, RocketCEA, CoolProp
Programming & Tools: Python (NumPy, SciPy), MATLAB, C, SQL, Git, Linux, LaTeX

Interests

Basketball — Sketch & Spray-Paint Art — Trekking & Outdoor Exploration